



Maastricht University

Department of Data Science and Knowledge Engineering

MATSE: Operations Research

Lecture 6. Markov Chains

[based on notes by: Frank Thuijsman and Steven Chaplick]

1856 –1922

Markov Chains

Given the present, the future is independent of the past.



1856 –1922



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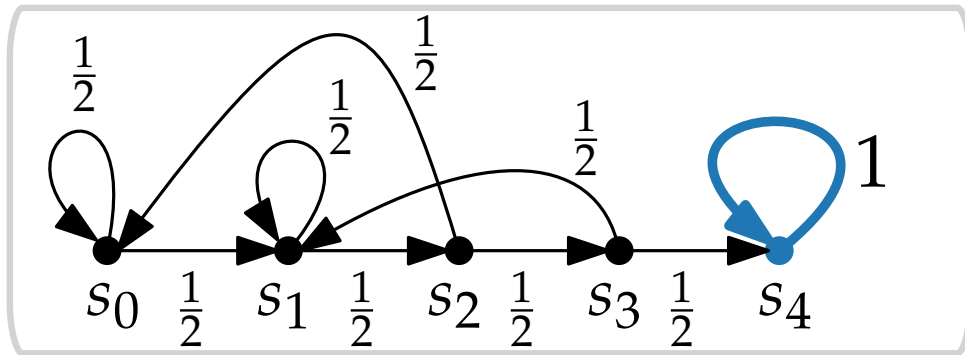


FIGURE 1.1. A randomly jumping frog. Whenever he tosses heads, he jumps to the other lily pad.

Levin, David A., and Yuval Peres. Markov chains and mixing times. Vol. 107. American Mathematical Soc., 2017.

Markov Chains

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- *states:* s_0, s_1, s_2, s_3, s_4

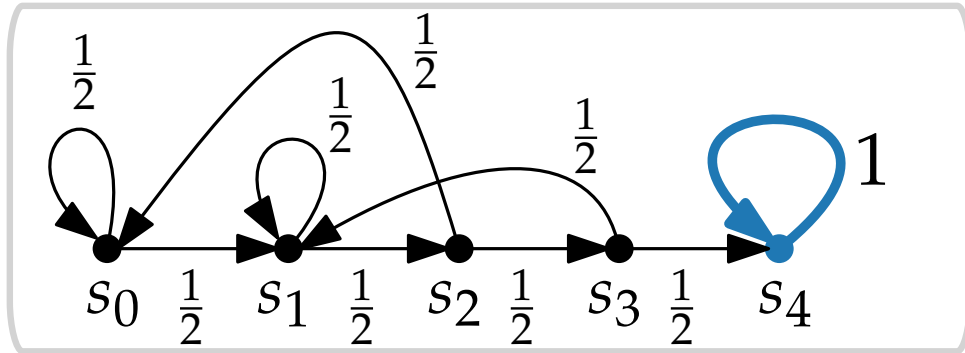
1856 – 1922

2 - 3



Markov Chains

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1856 – 1922



Transition Matrix

$$\begin{array}{c} \text{from} \\ s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{array} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = P$$

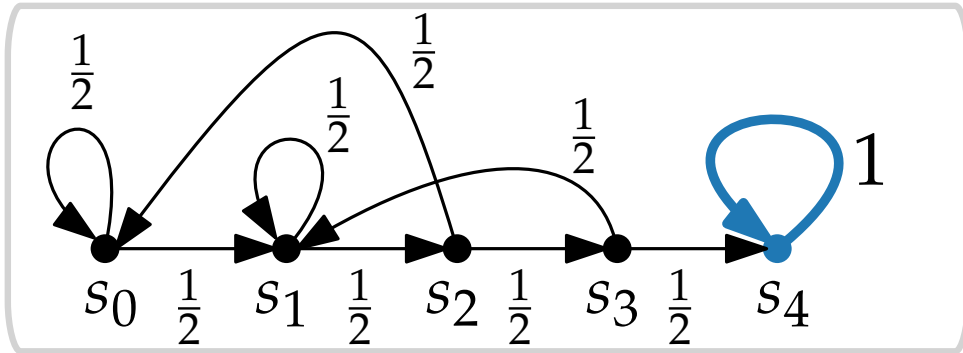
to
 $s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

1856 – 1922



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- Note: sum of any row of $P = 1$

from

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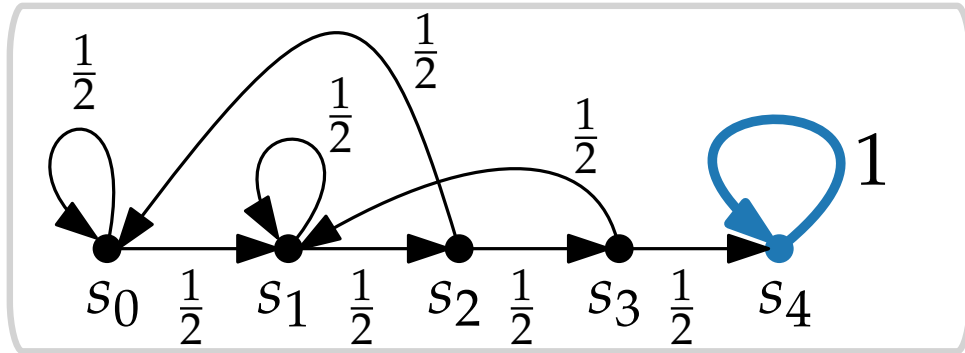
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only visited **finitely often**

1856 – 1922



from

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 s_0 \\
 s_1 \\
 s_2 \\
 s_3 \\
 s_4
 \end{matrix}
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 \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\
 \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\
 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\
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 = P$$

to

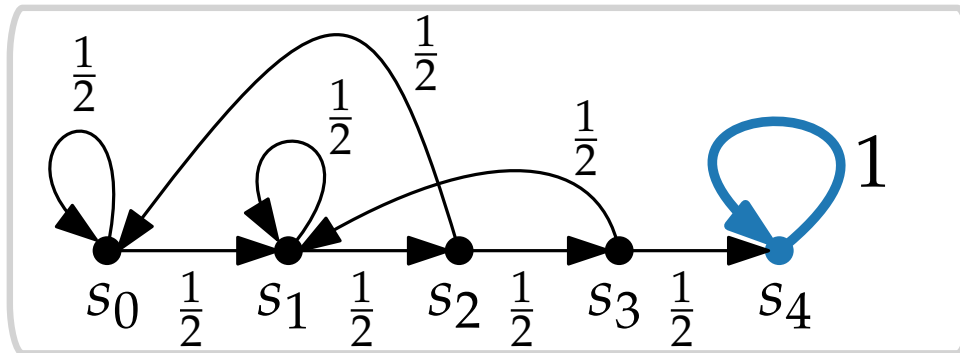
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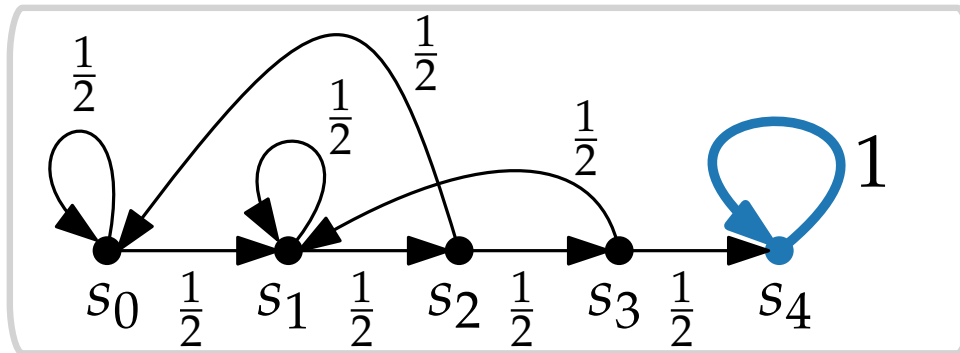
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- s_0, s_1, s_2, s_3 are *transient states* :
only visited **finitely often**
- s_4 is a *recurrent state*: it is visited infinitely often.
- s_4 is also an **absorbing state** : it is inescapable.

from

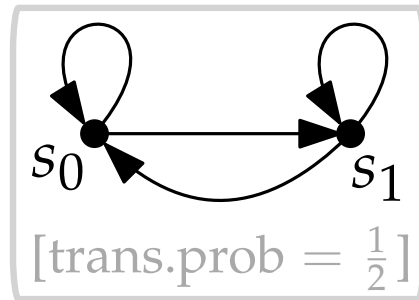
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Recurrent vs Absorbing

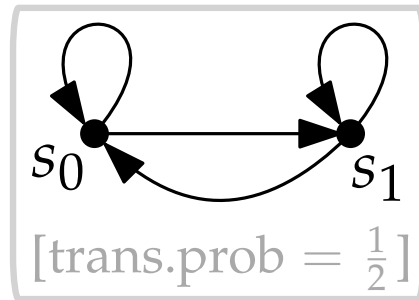
Consider the Markov chain



$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Recurrent vs Absorbing

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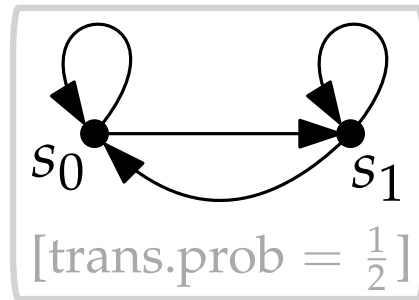


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- All states are recurrent
- None of them is absorbing

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- All states are recurrent
- None of them is absorbing
- Absorbing $\Rightarrow$ Recurrent
The vice-versa is not true.

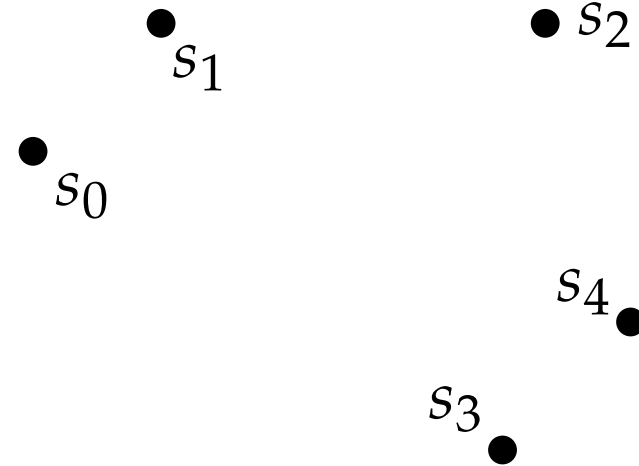
A Bigger Example

from \ to	s_0	s_1	s_2	s_3	s_4	
s_0	0.6	0.3	0	0.1	0	:
s_1	0.4	0.4	0.1	0	0.1	
s_2	0	0	1	0	0	
s_3	0	0	0	0.4	0.6	
s_4	0	0	0	0.3	0.7	

A Bigger Example

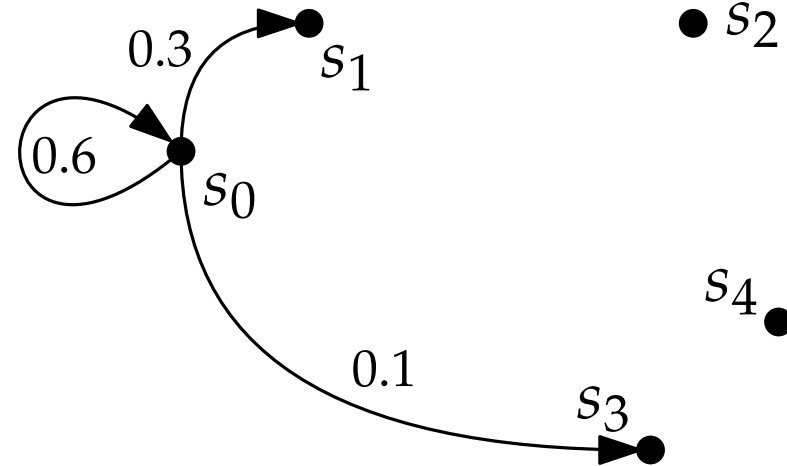
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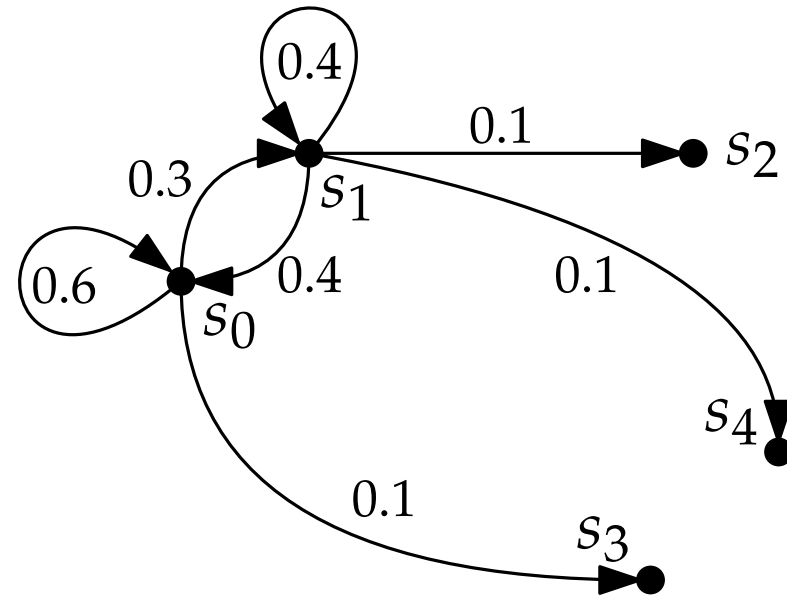
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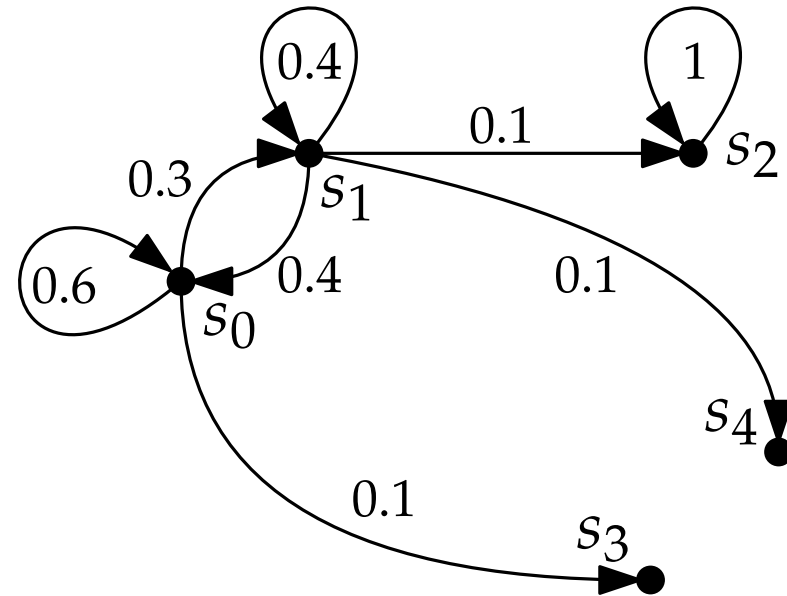
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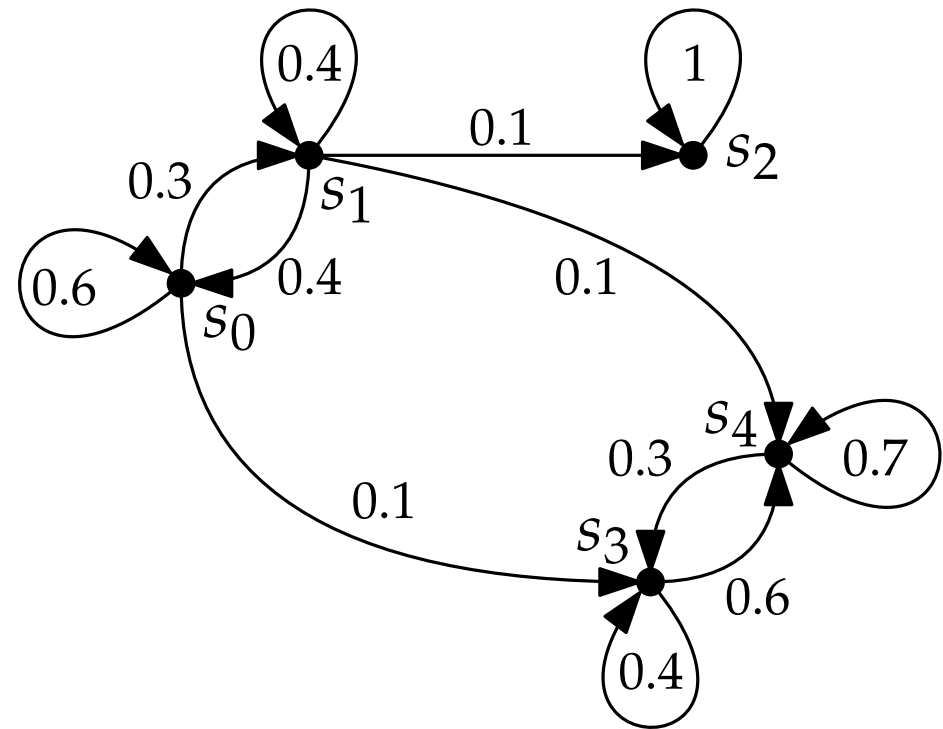
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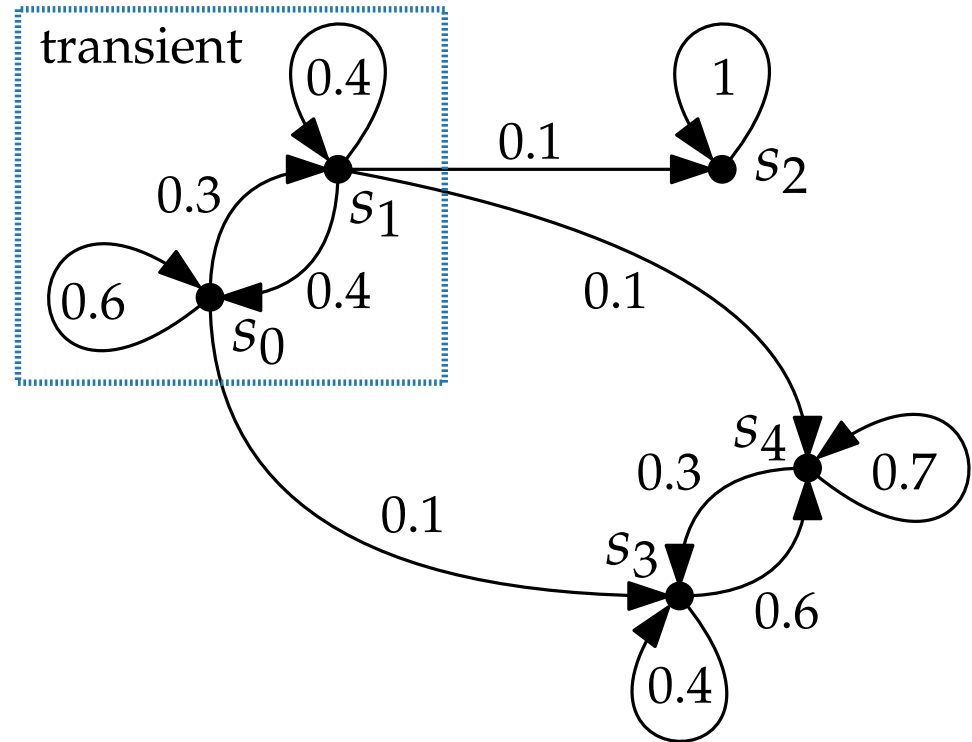
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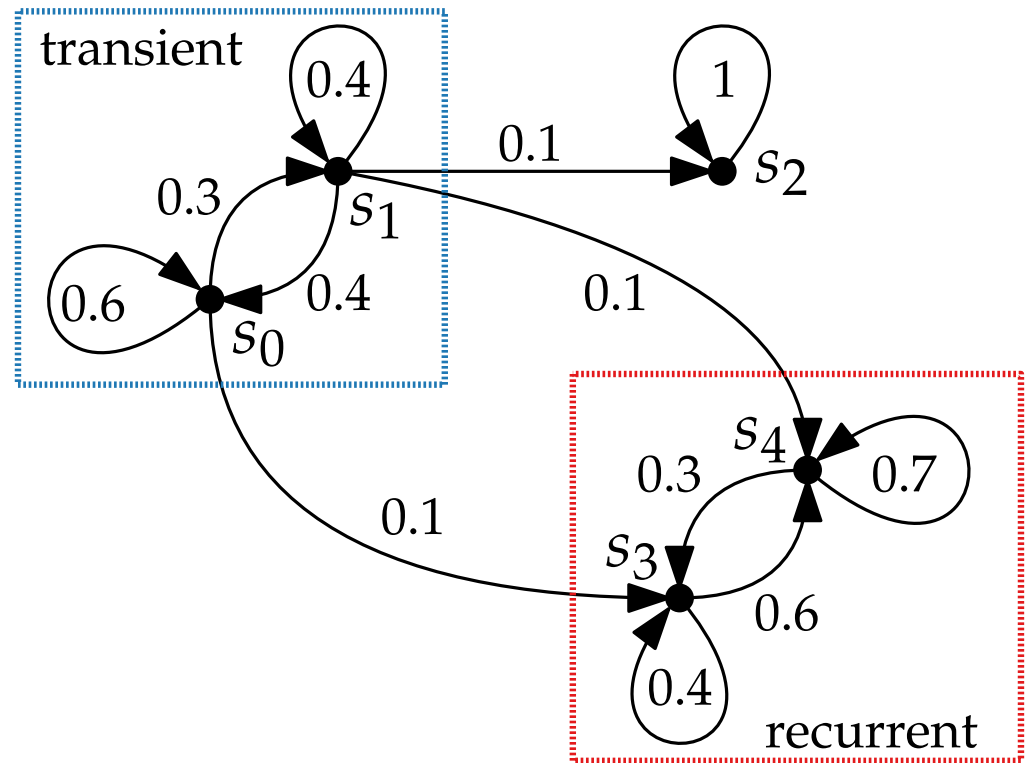
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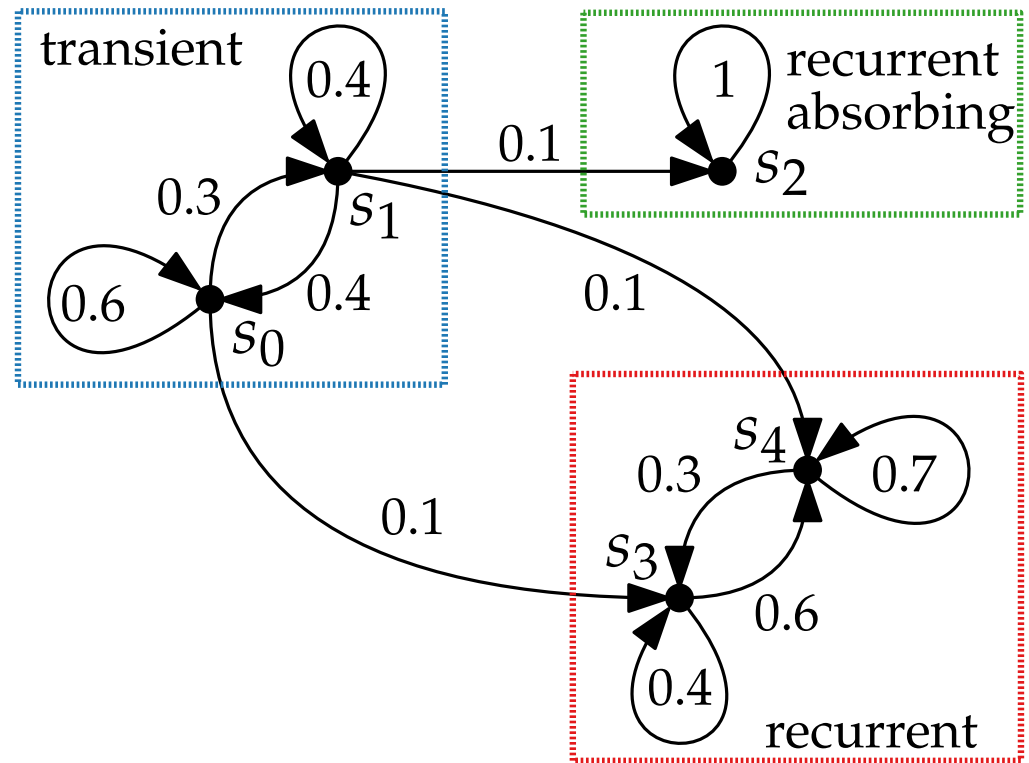
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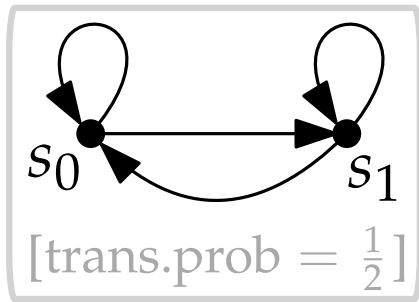


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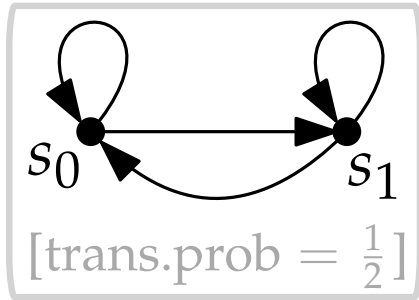
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Long Run Functions: π_i

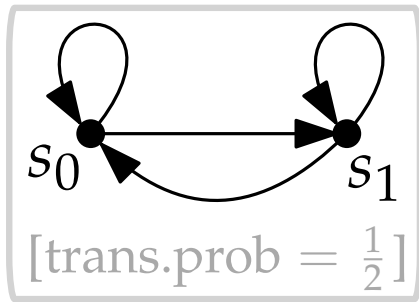


Long Run Functions: π_i



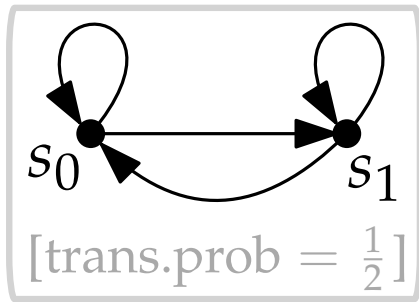
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Long Run Functions: π_i



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- *irreducible* $\coloneqq$ **all** pairs of states communicate.

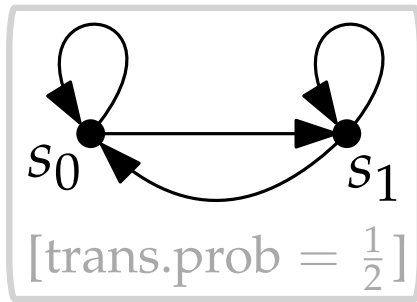
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Obs: irreducible $\Rightarrow$ all states are recurrent

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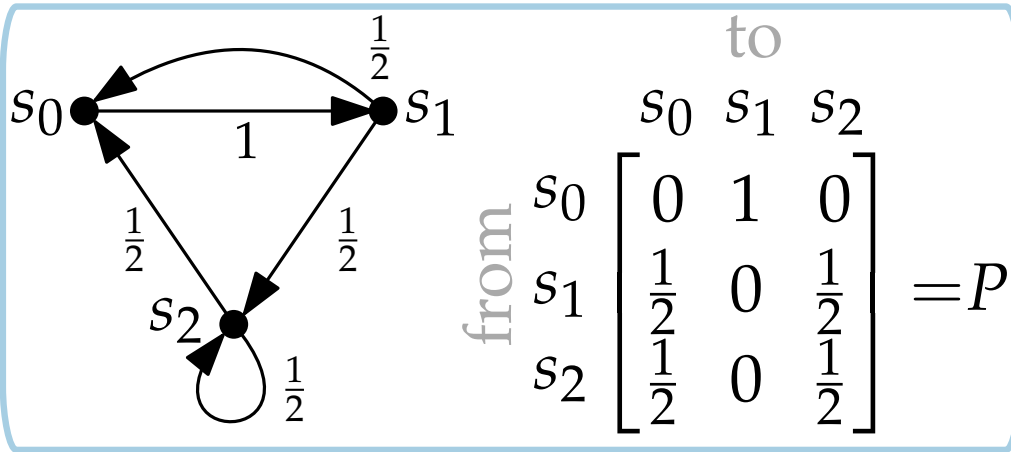
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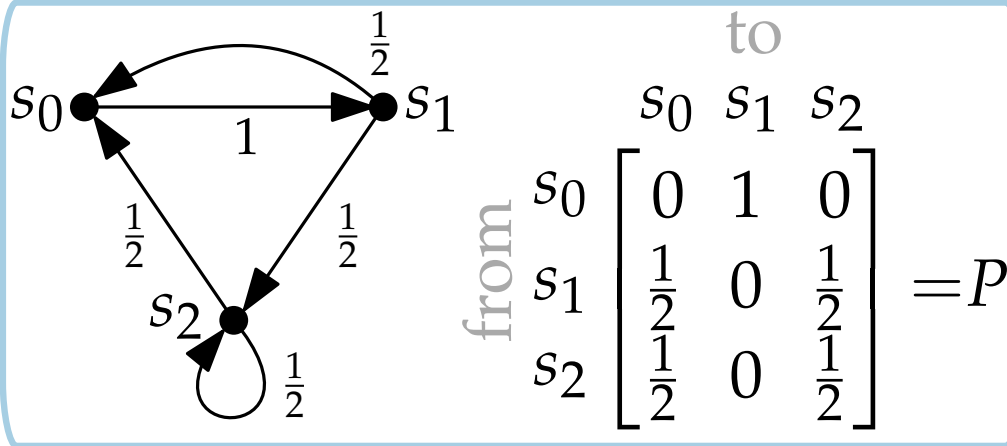
- *steady-state probabilities*: $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots, \pi_k)$

For a Markov chain, π_i is the fraction all time spent in state s_i .

Finding Steady-State Probabilities

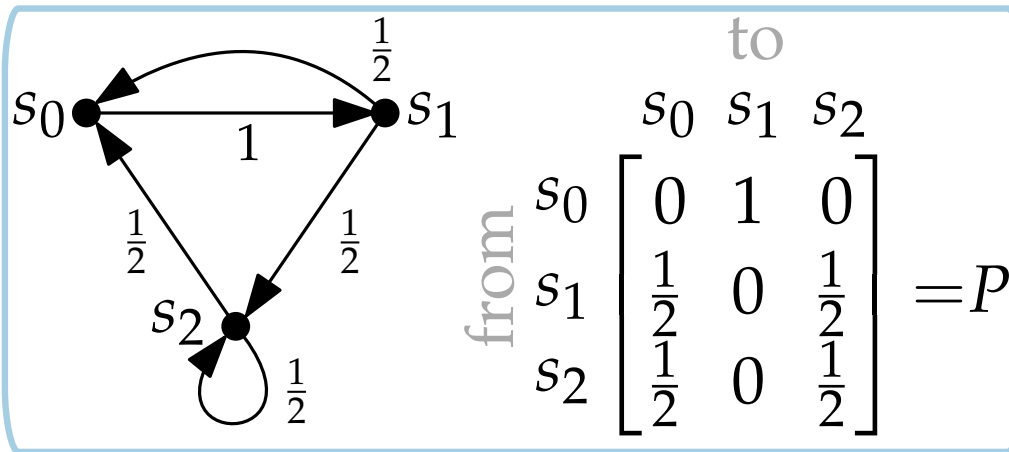


Finding Steady-State Probabilities



How to calculate $\pi = (\pi_0, \pi_1, \pi_2)$?

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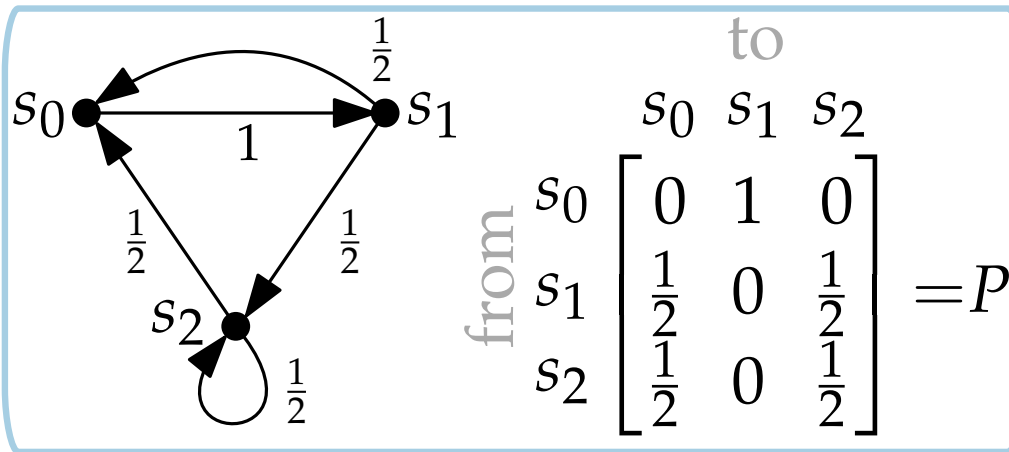
What properties hold for π ?

[when all pairs communicate]

$$\sum_i \pi_i = 1, \text{ and } \pi P = \pi$$

[Prob(being at s_i) = Prob(getting to s_i)
= Prob(via **incoming** edges of s_i)]

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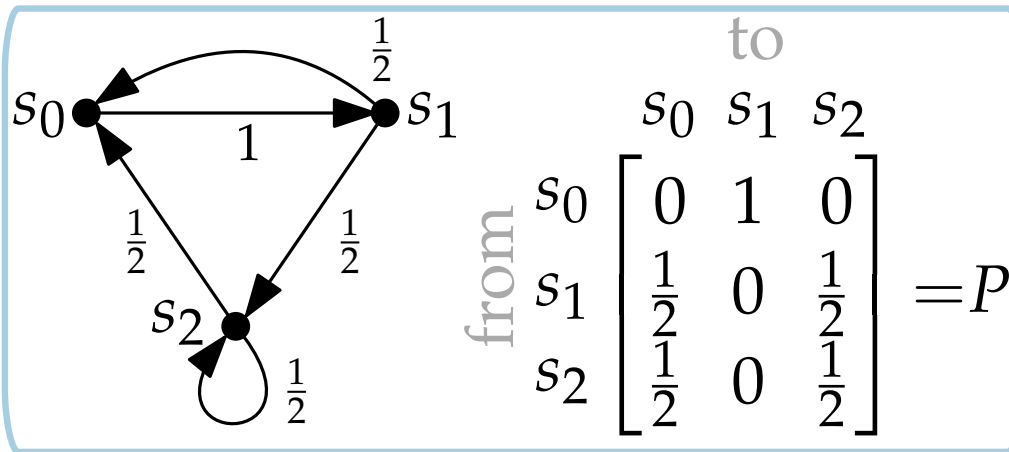
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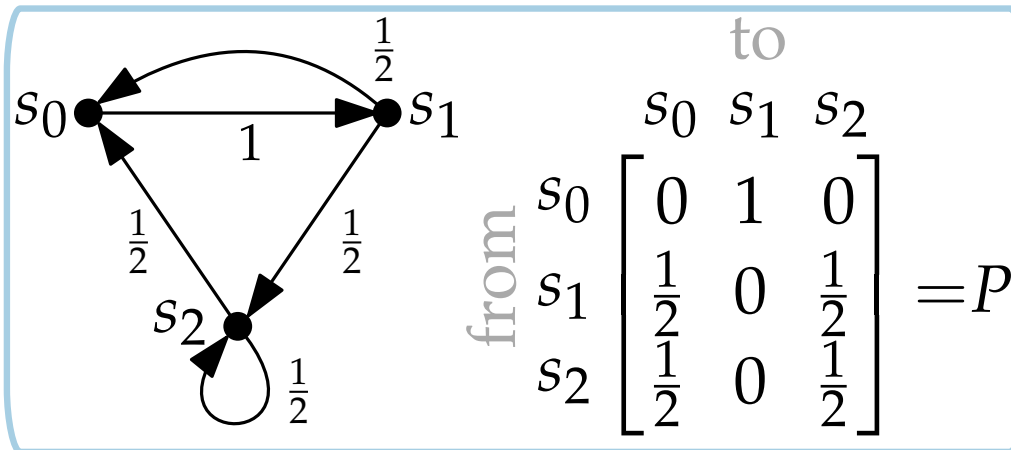
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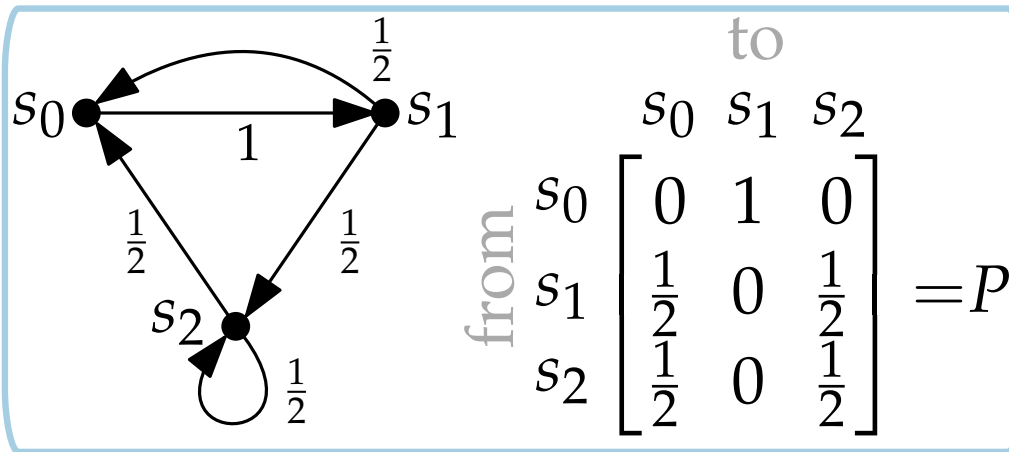
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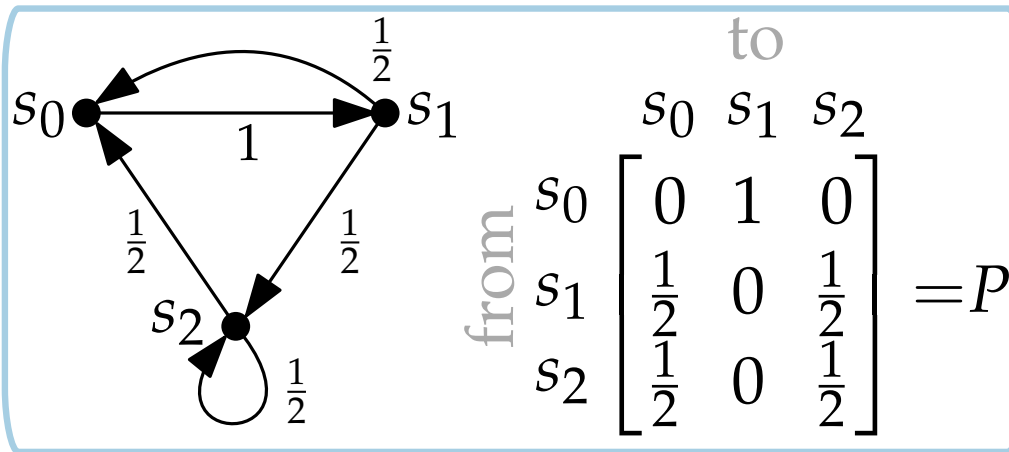
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$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

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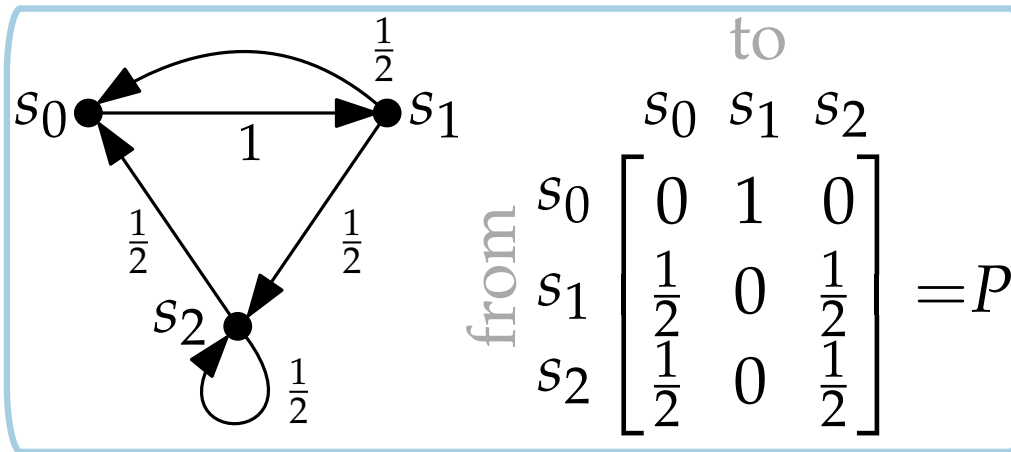
Fraction In = Fraction Out

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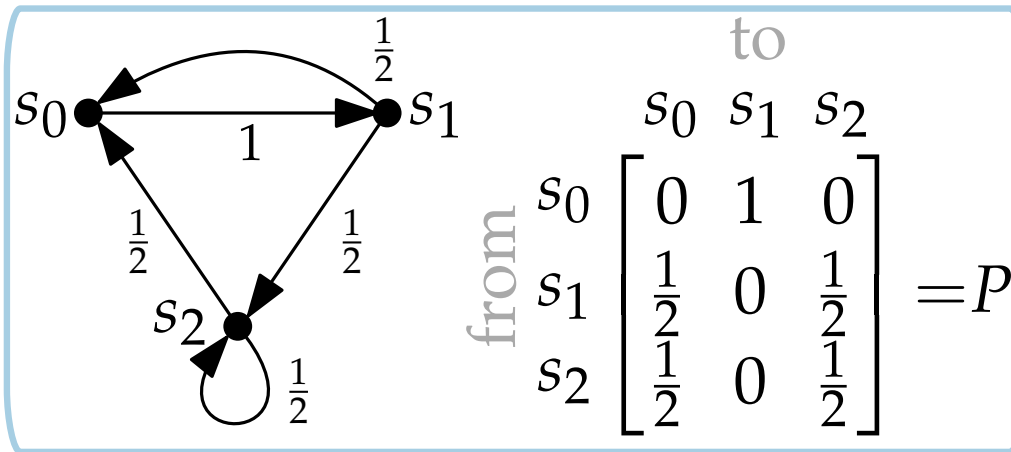
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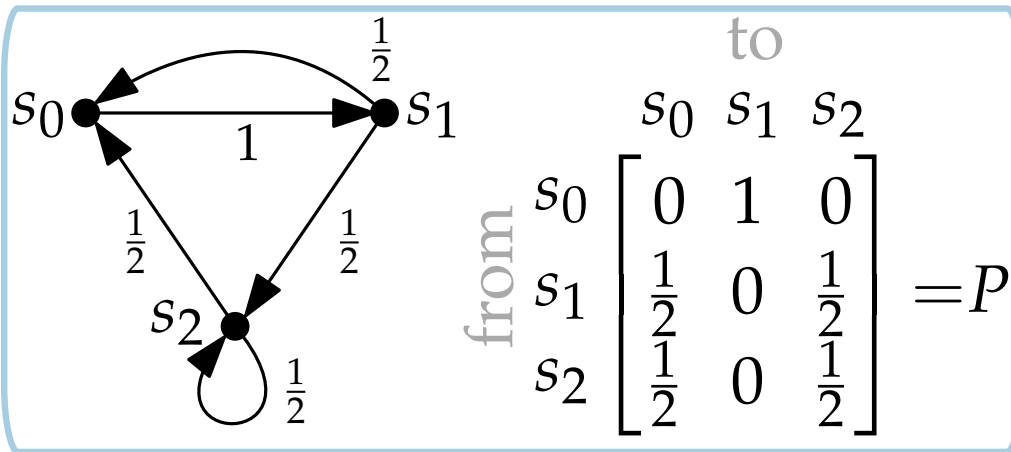
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$$\frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 = \frac{1}{2}\pi_2 + \frac{1}{2}\pi_2$$

$$\begin{cases} \pi_1 = \pi_2 \\ \pi_0 = \pi_2 \\ \pi_0 + \pi_1 + \pi_2 = 1 \end{cases}$$

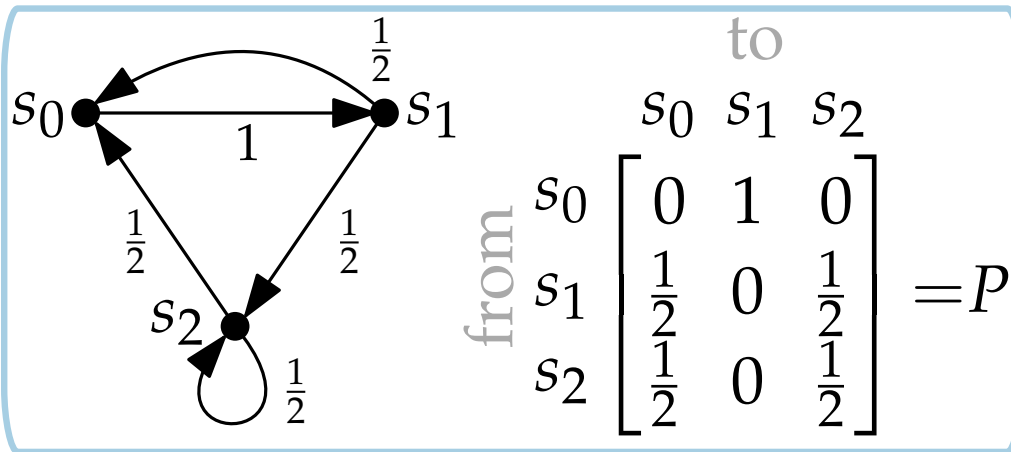
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“Short” run?



$P :=$ transition matrix, **1-step**
transition probabilities

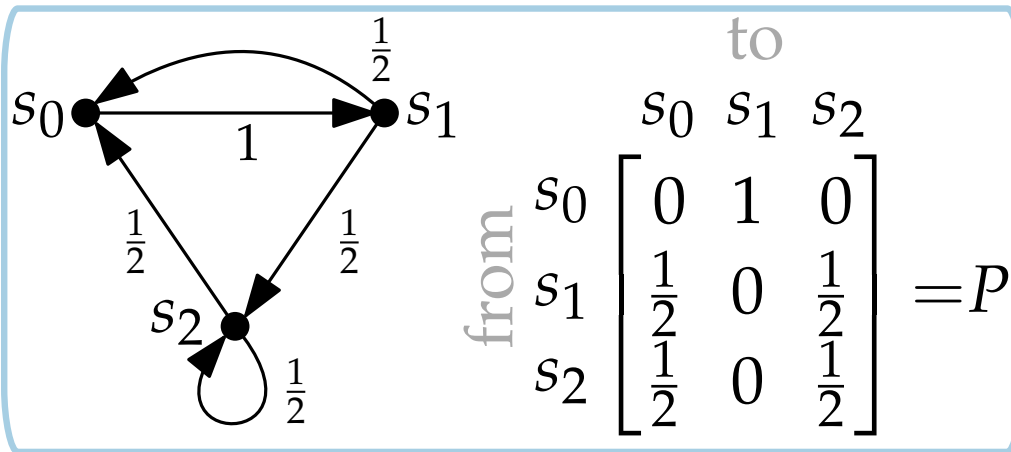
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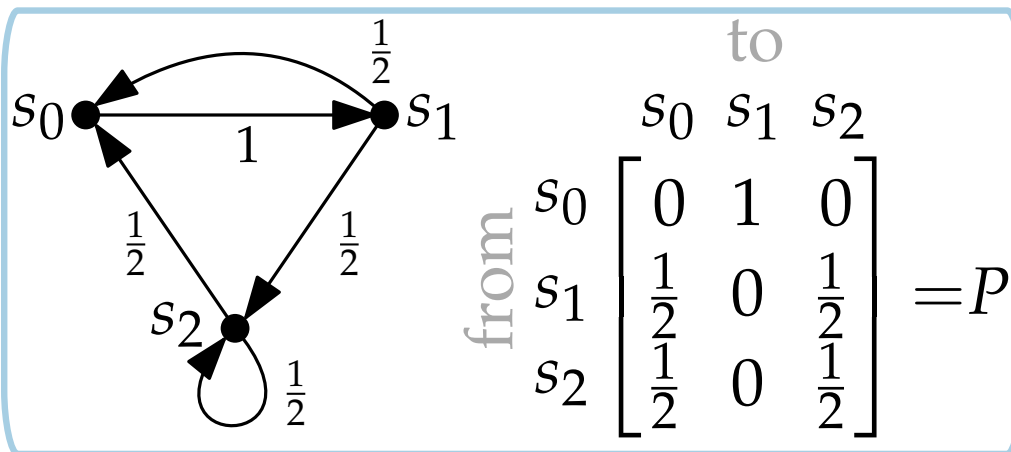


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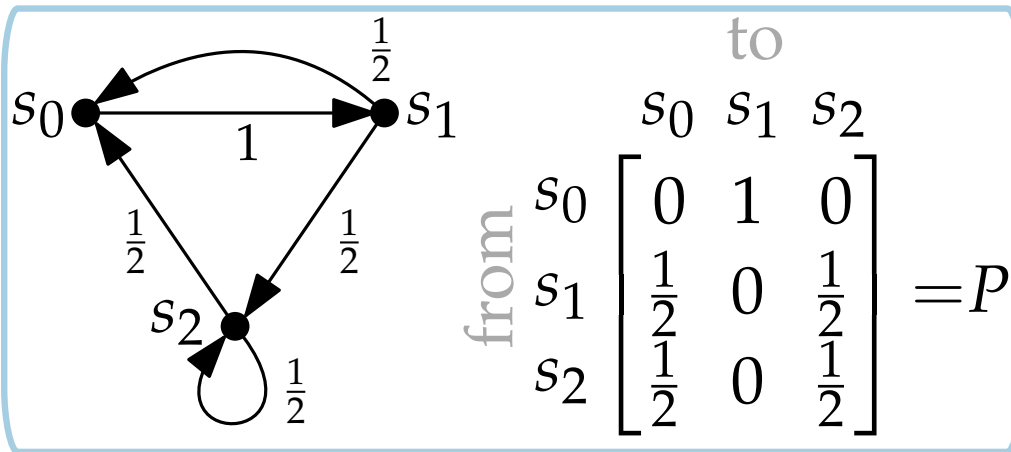
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$P_{ij}^{(2)}$ = probability we go from i to *somewhere* and then to j

$$= \sum_b \text{Prob}(i \rightarrow b \rightarrow j) = \sum_b P_{ib} P_{bj}$$

“Short” run?



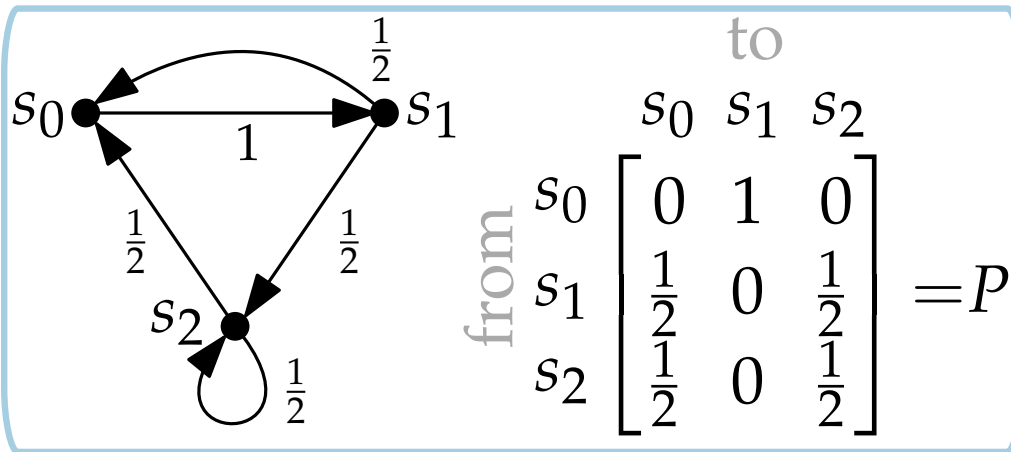
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 P_{ij}^{(2)} &= \text{probability we go from } i \text{ to somewhere and then to } j \\
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Therefore, $P^{(2)} = PP = P^2 =$

$$\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

"Longer" run

What about $P^{(t)}$?

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$$\begin{aligned} P_{ij}^{(t)} &= \sum_{b=0}^k \text{Prob}(i \xrightarrow[t \text{ steps}]{t-1} b \rightarrow j) = \sum_{b=0}^k P_{ib}^{(t-1)} P_{bj} \\ &= \sum_{b=0}^k P_{ib}^{t-1} P_{bj} \text{ [by induction]} \end{aligned}$$

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Can we relate long runs π to short runs $P^{(t)}$?

Maybe they relate in the **limit**?

“Long”run vs. “short” run

$P^t = P P P \dots P = t\text{-step transition matrix}$

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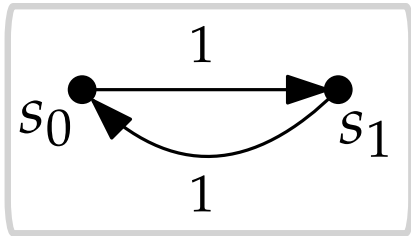
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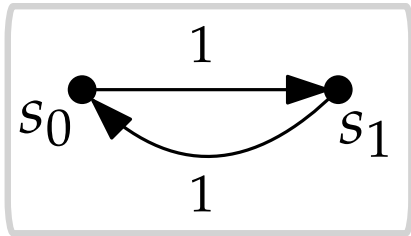


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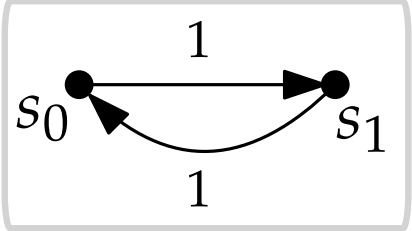
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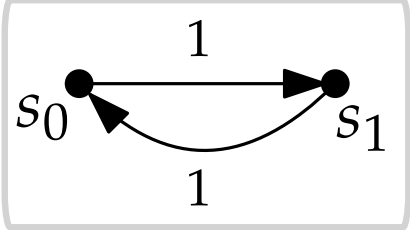
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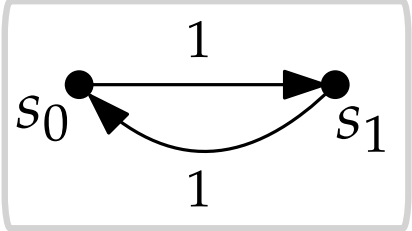
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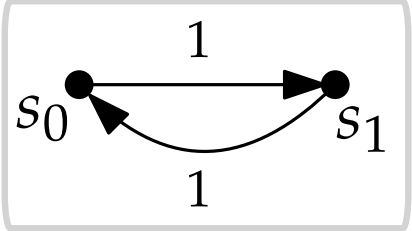
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$\rightsquigarrow$ No such limit for this *periodic* Markov chain!

Periodicity Properties

- **Period** of state i : the integer number t such that $p_{ii}^{(n)} = 0$ for all n but $t, 2t, 3t, \dots$

$$t = \gcd\{n \in \mathbb{N} \text{ s.t. } p_{ii}^{(n)} > 0\}$$

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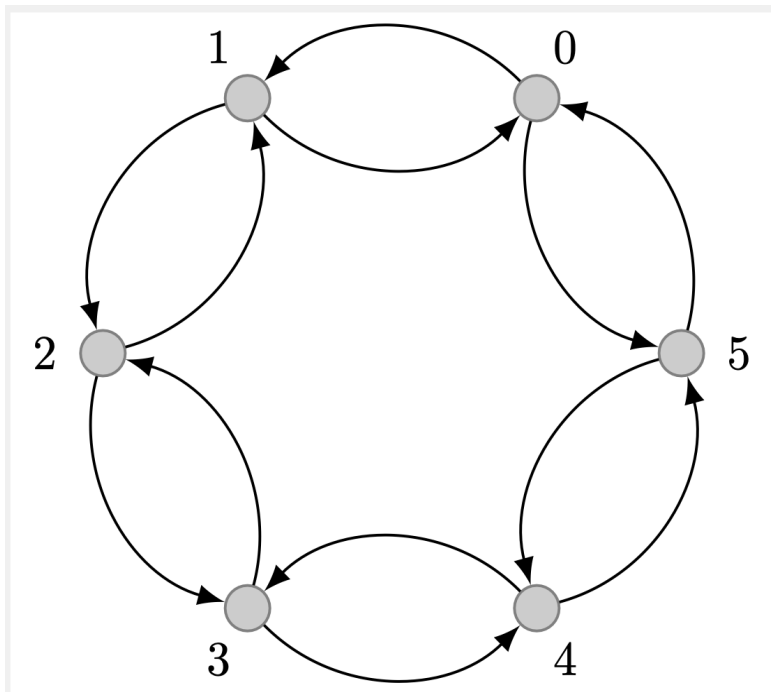
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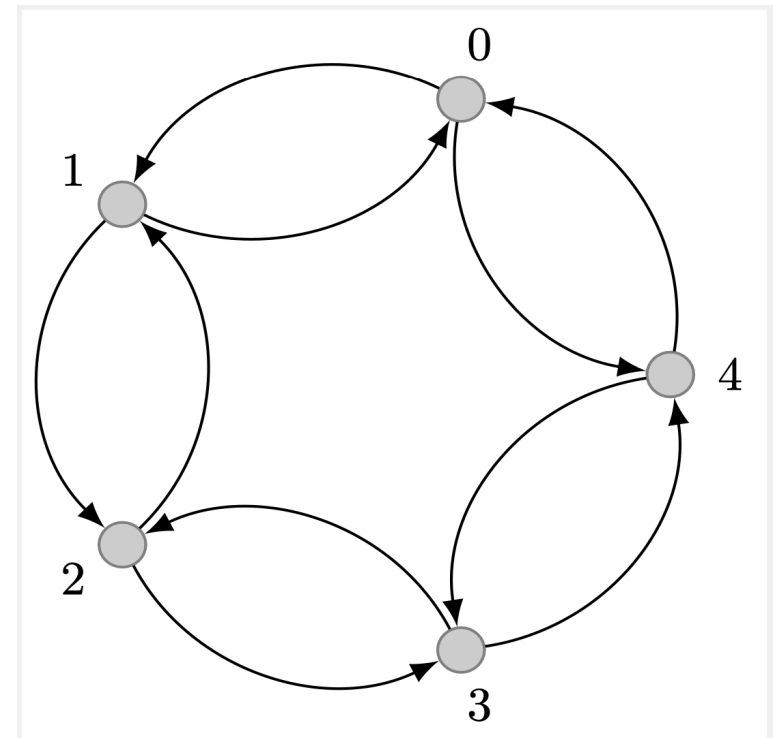
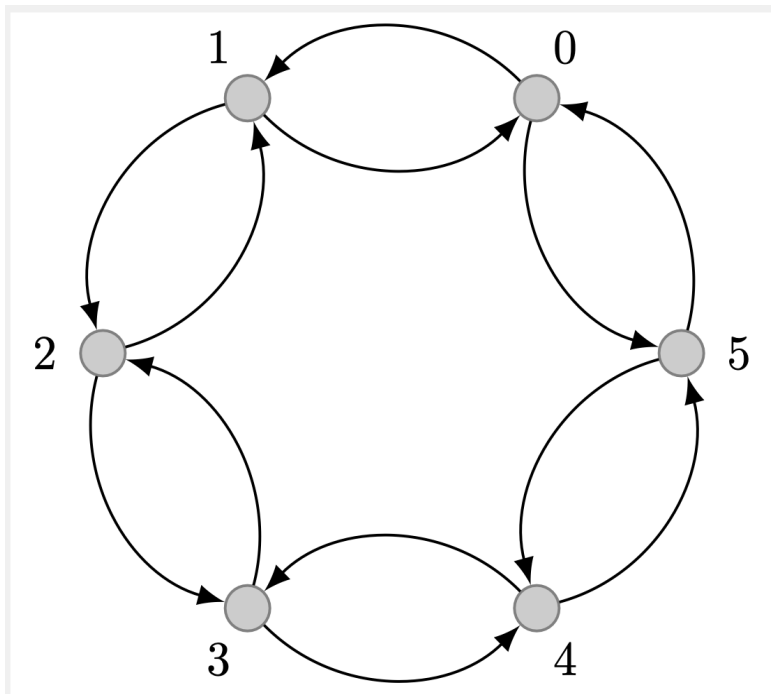


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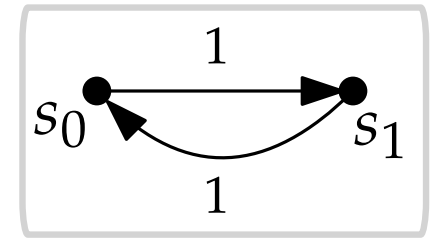
“Long” run vs. “short” run

$$Q_{ij} = \lim_{t \rightarrow \infty} P_{ij}^t = \pi_j$$

True if the Markov chain is *Aperiodic* and *Irreducible*

“Long” run vs. “short” run

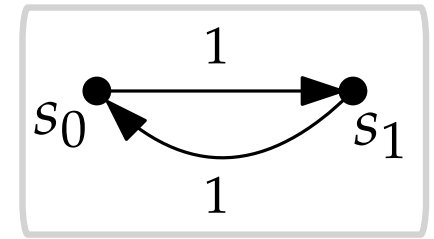
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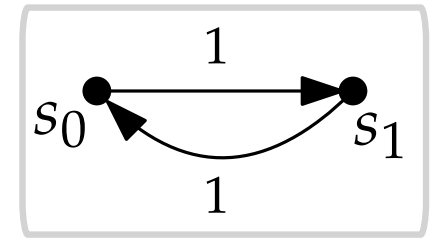
$Q_{ij}^{(t)} :=$ avg. time (over t -steps) in state j , starting in state i

$$Q^{(t)} = \frac{1}{t} \left(I + P^1 + P^2 + \dots + P^{t-1} \right) [I := \text{identity matrix}]$$

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row(i) of Q is π for starting state i .

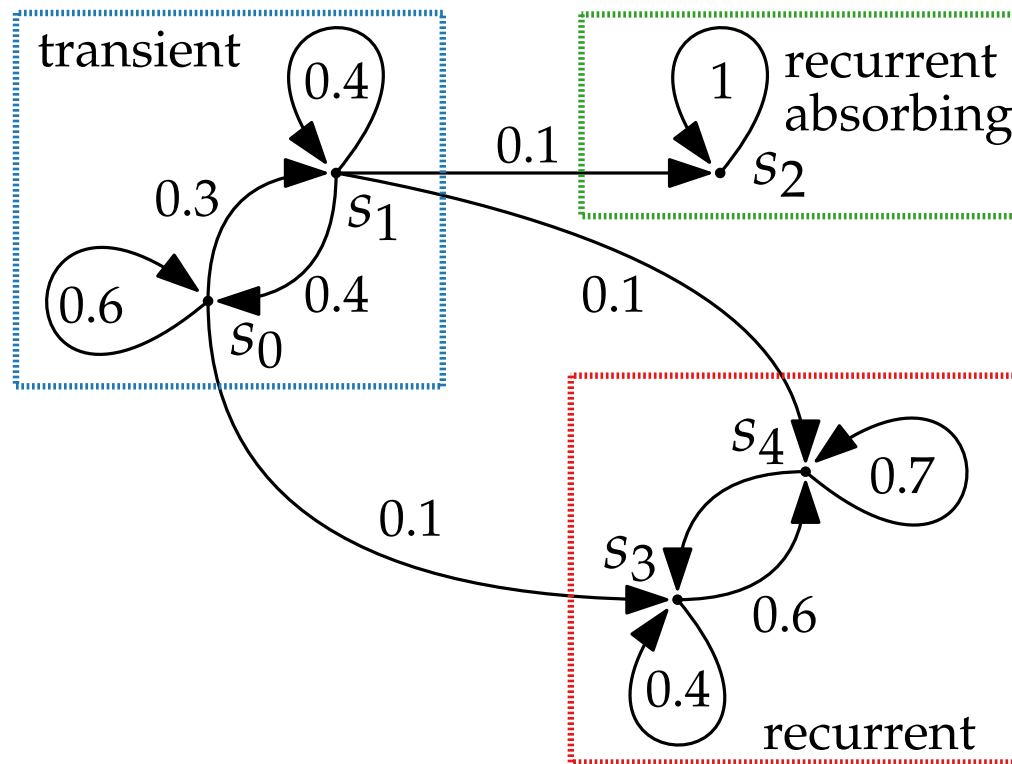
Irreducible $\Rightarrow$ each row of Q will be the same! e.g. $\pi_j = Q_{ij}$

Example

	from \ to	s_0	s_1	s_2	s_3	s_4
s_0		0.6	0.3	0	0.1	0
s_1		0.4	0.4	0.1	0	0.1
s_2		0	0	1	0	0
s_3		0	0	0	0.4	0.6
s_4		0	0	0	0.3	0.7

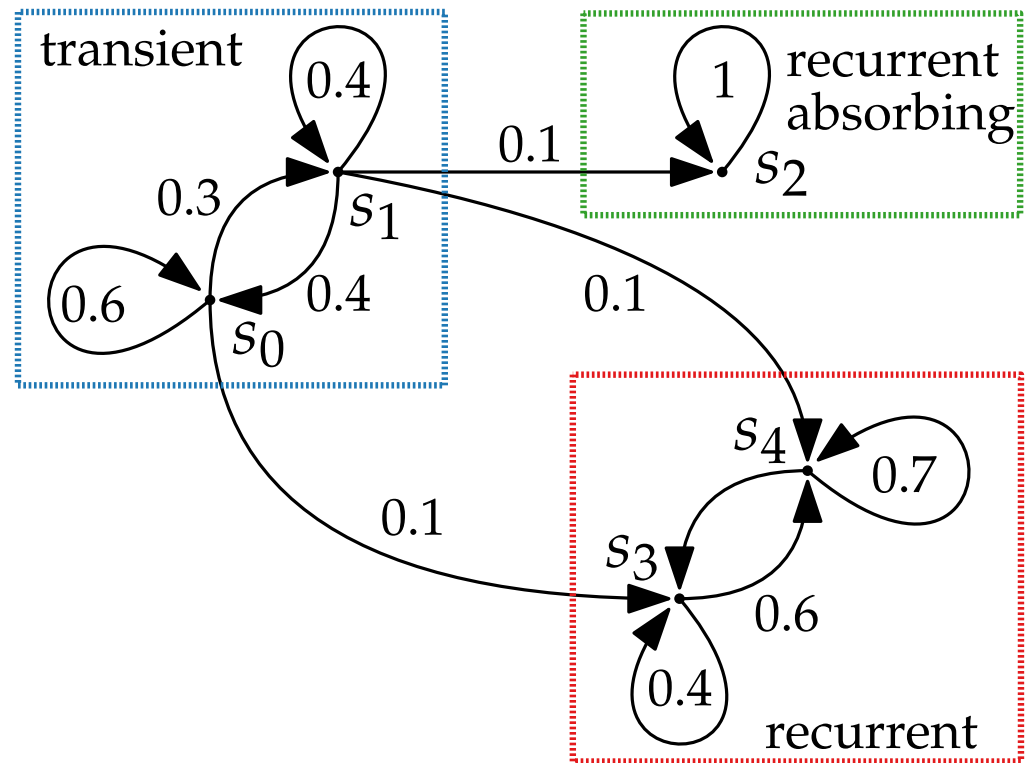
Starting in s_0 :

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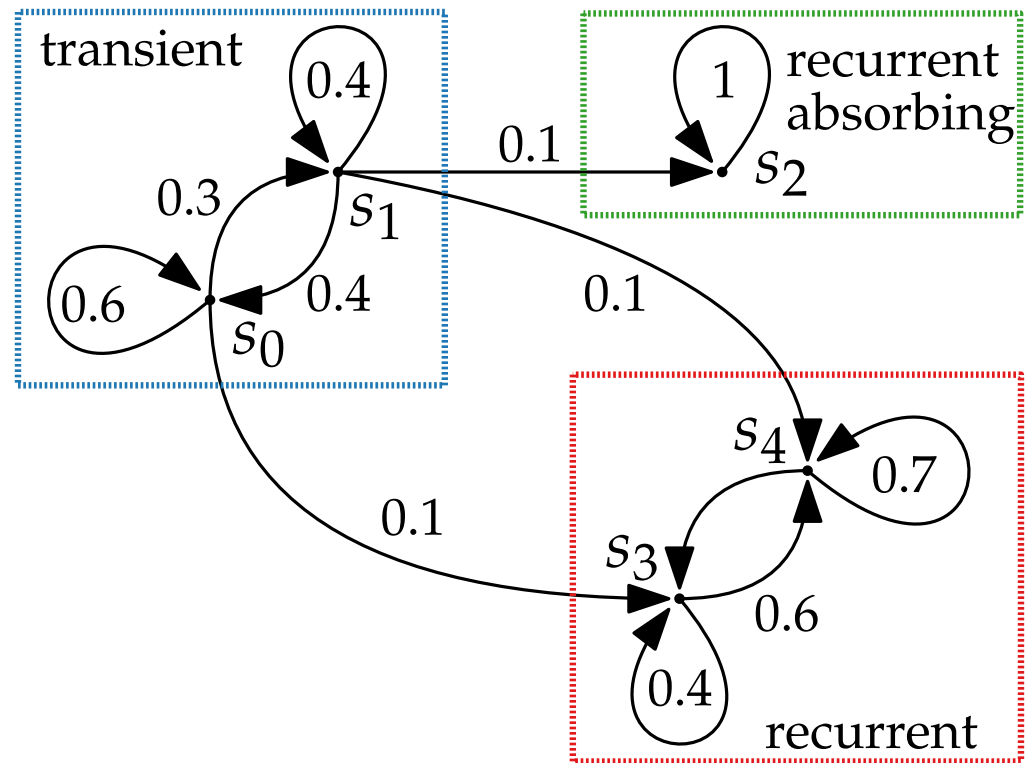
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Let P_x be the probability of reaching s_2 starting at x

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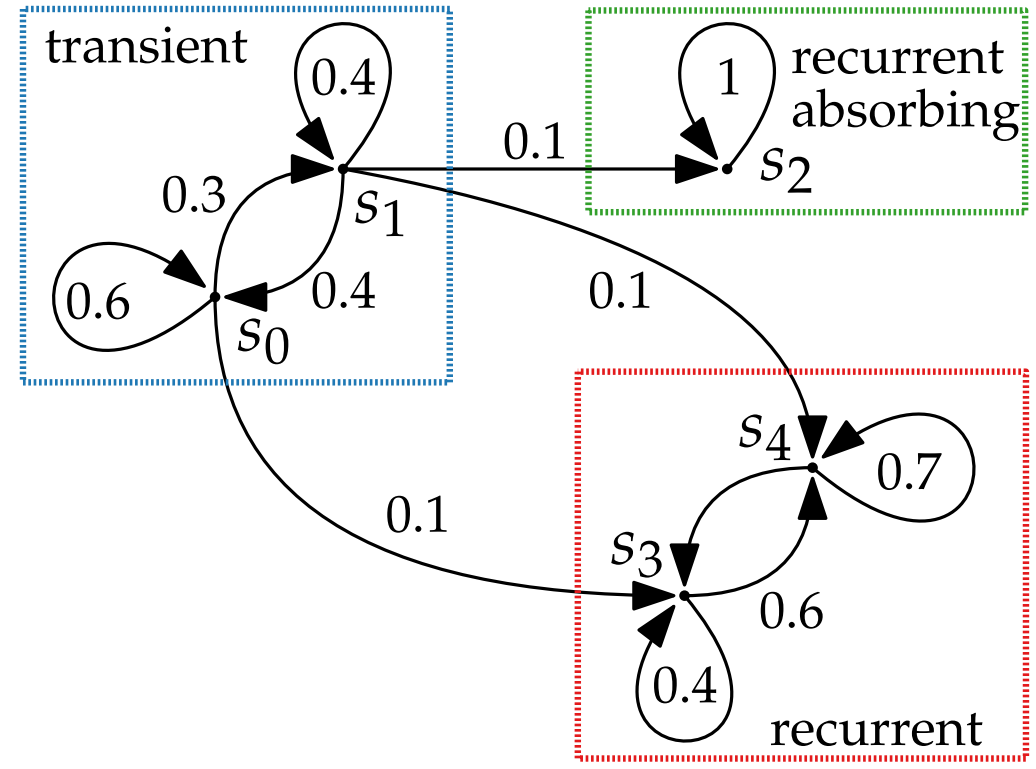


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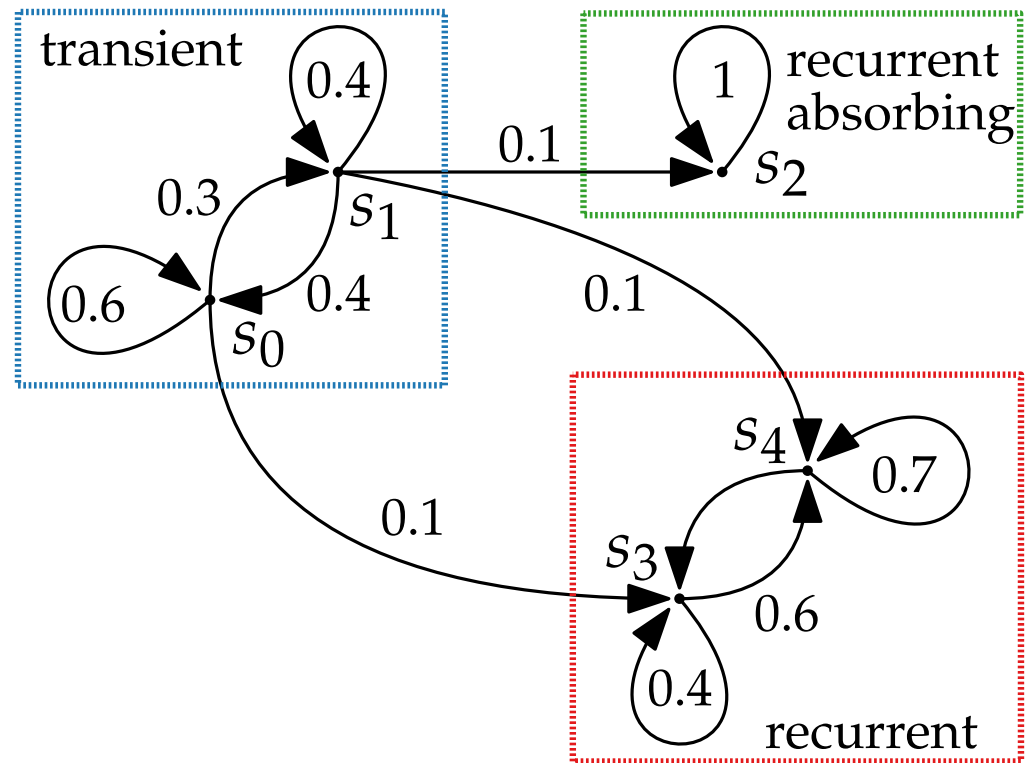


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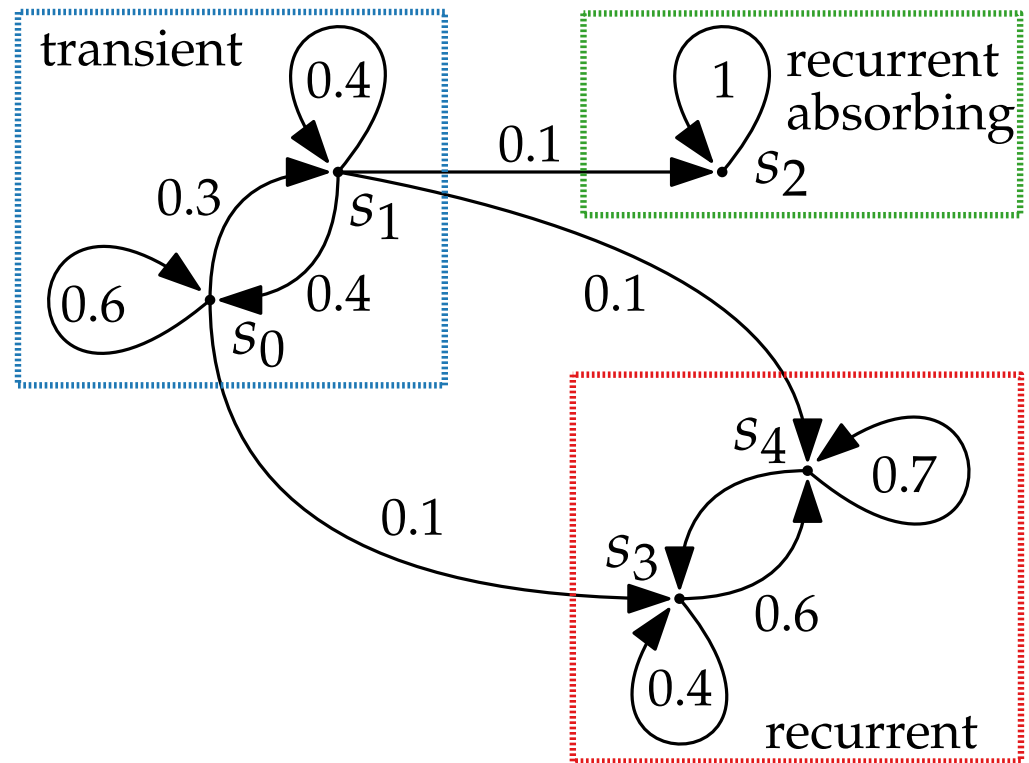
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$$\Rightarrow P_{0,3} = P_{0,4} = 1 - P_{0,2} = 1 - \frac{1}{4} = \frac{3}{4}$$

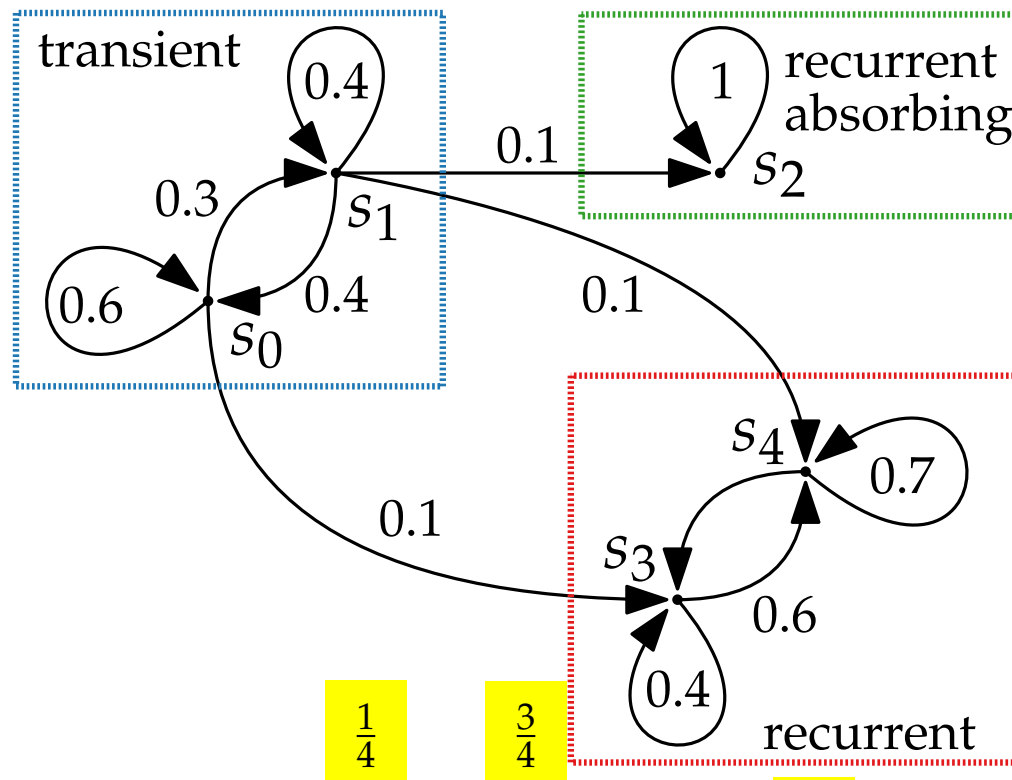
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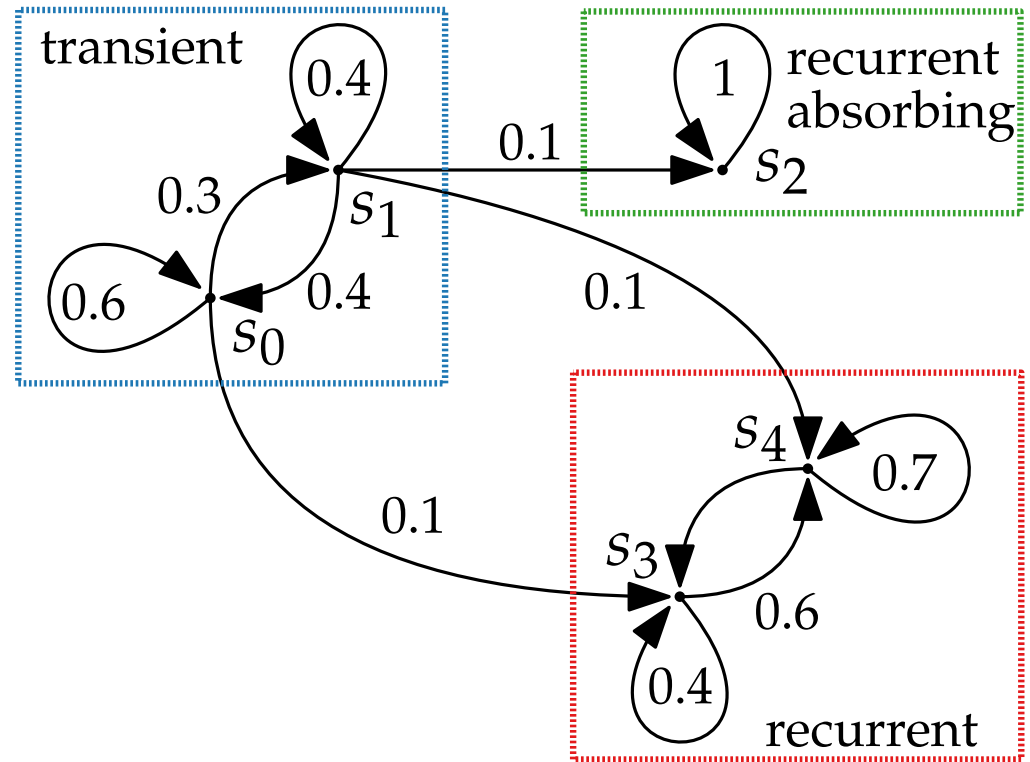
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to

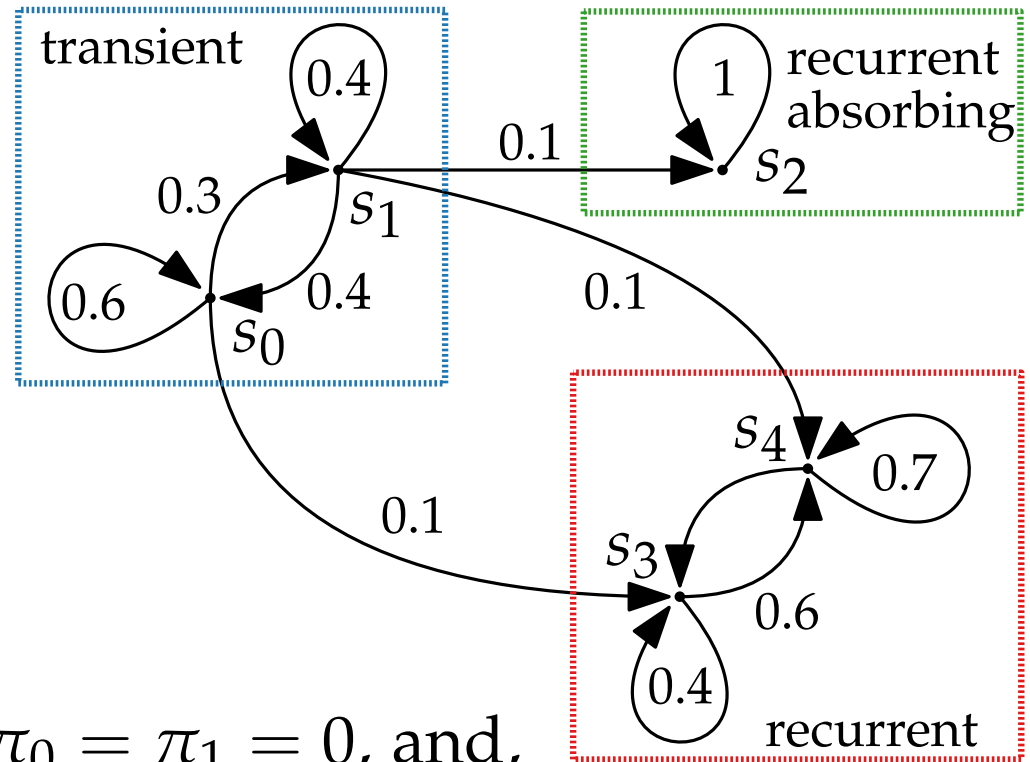
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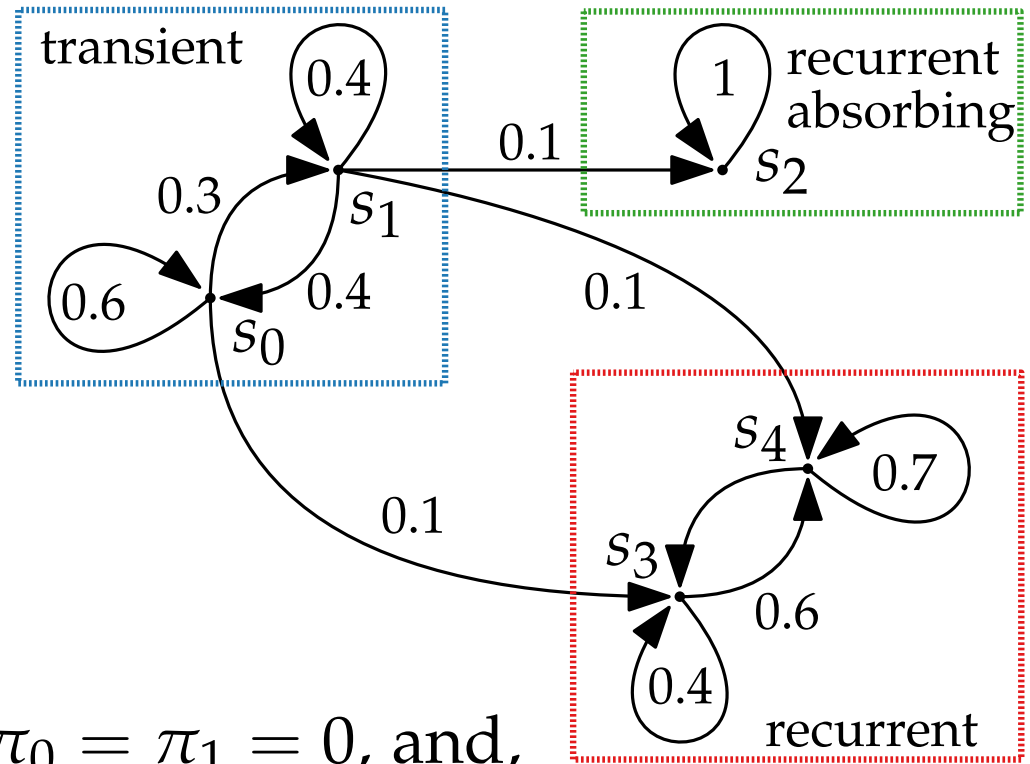


Since s_0 and s_1 are transient, $\pi_0 = \pi_1 = 0$, and, since s_2 is absorbing, from a)(i), $\pi_2 = \frac{1}{4}$.

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to

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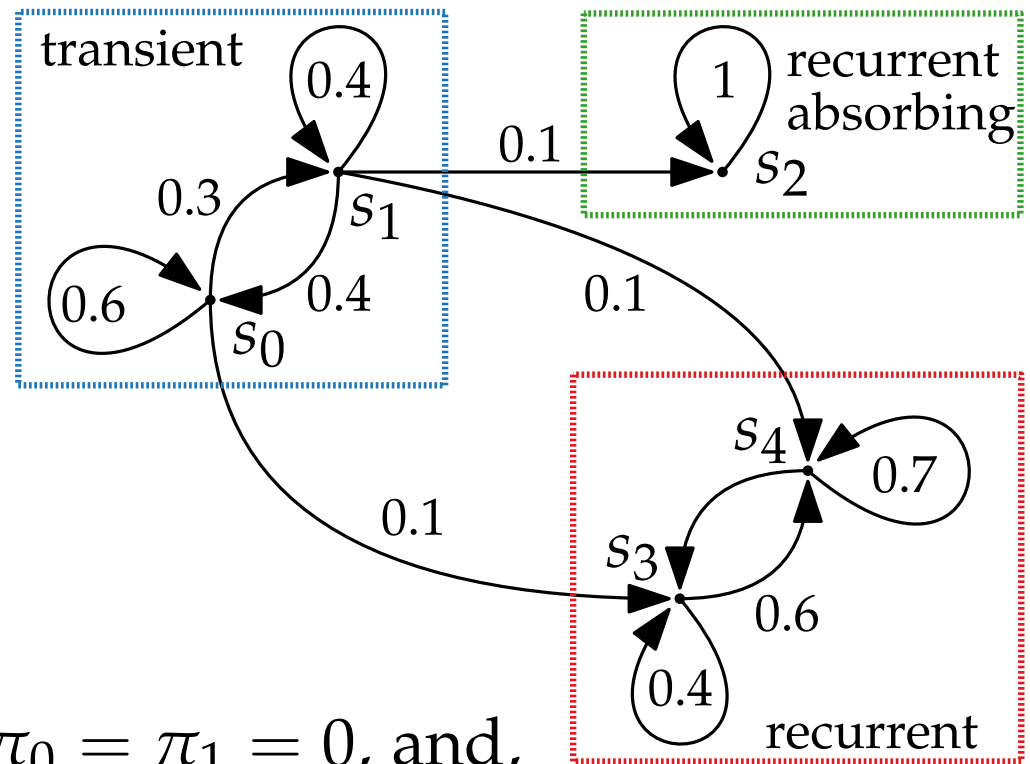
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Since s_3 and s_4 are in the same recurrent class and this class is **reached** with prob. $\frac{3}{4}$,

Example

to

	s_0	s_1	s_2	s_3	s_4
s_0	0.6	0.3	0	0.1	0
s_1	0.4	0.4	0.1	0	0.1
s_2	0	0	1	0	0
s_3	0	0	0	0.4	0.6
s_4	0	0	0	0.3	0.7



Since s_0 and s_1 are transient, $\pi_0 = \pi_1 = 0$, and, since s_2 is absorbing, from a)(i), $\pi_2 = \frac{1}{4}$.

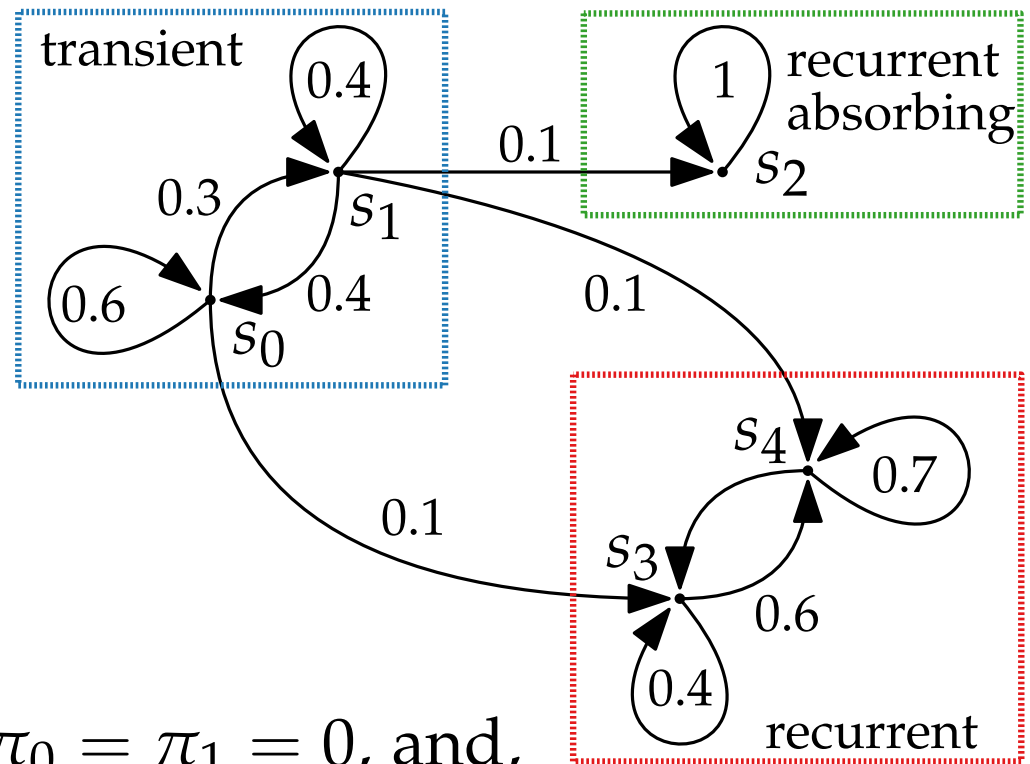
Since s_3 and s_4 are in the same recurrent class and this class is **reached** with prob. $\frac{3}{4}$,

$$(\pi_3, \pi_4) \begin{bmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{bmatrix} = (\pi_3, \pi_4) :$$

Example

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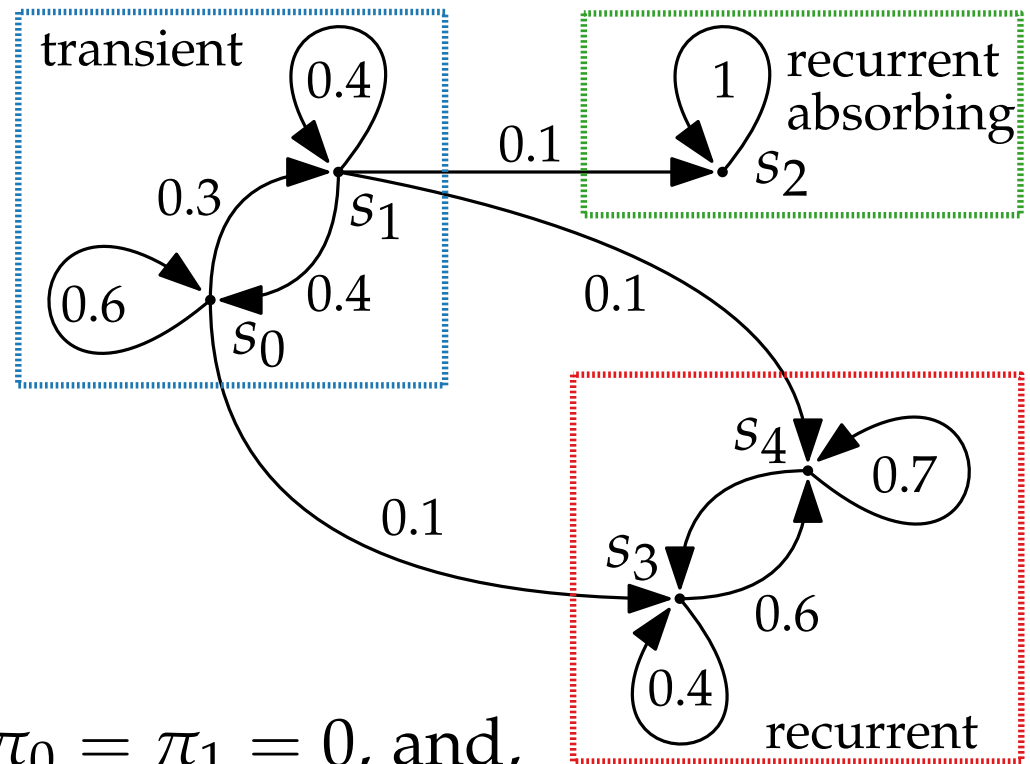
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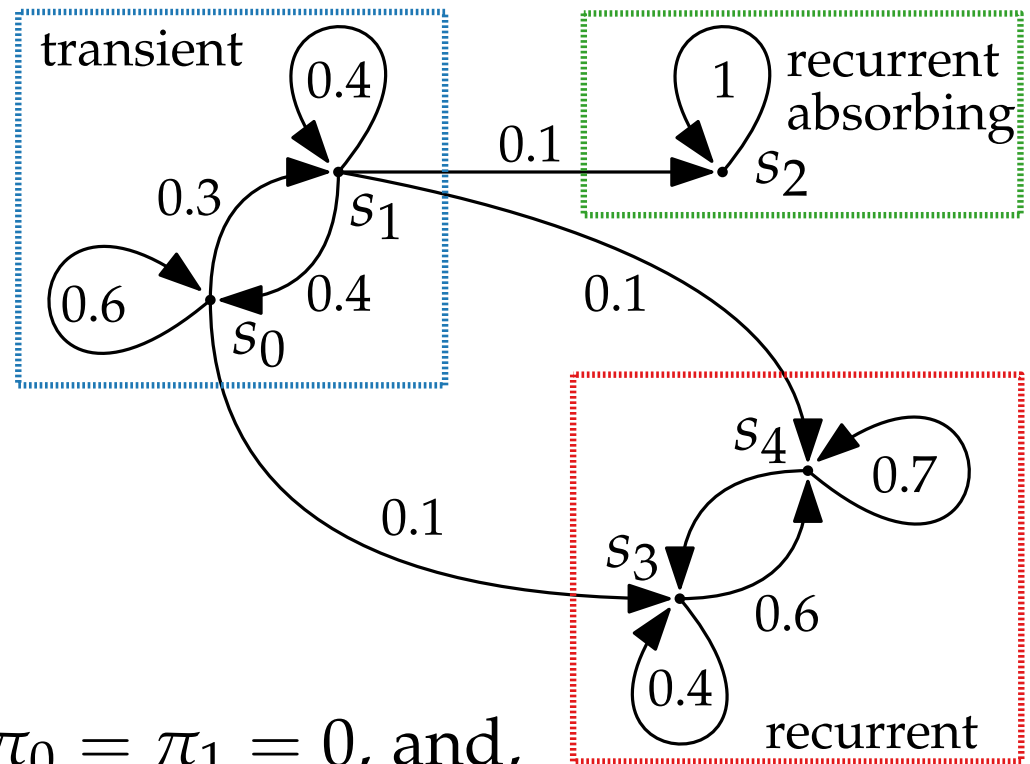
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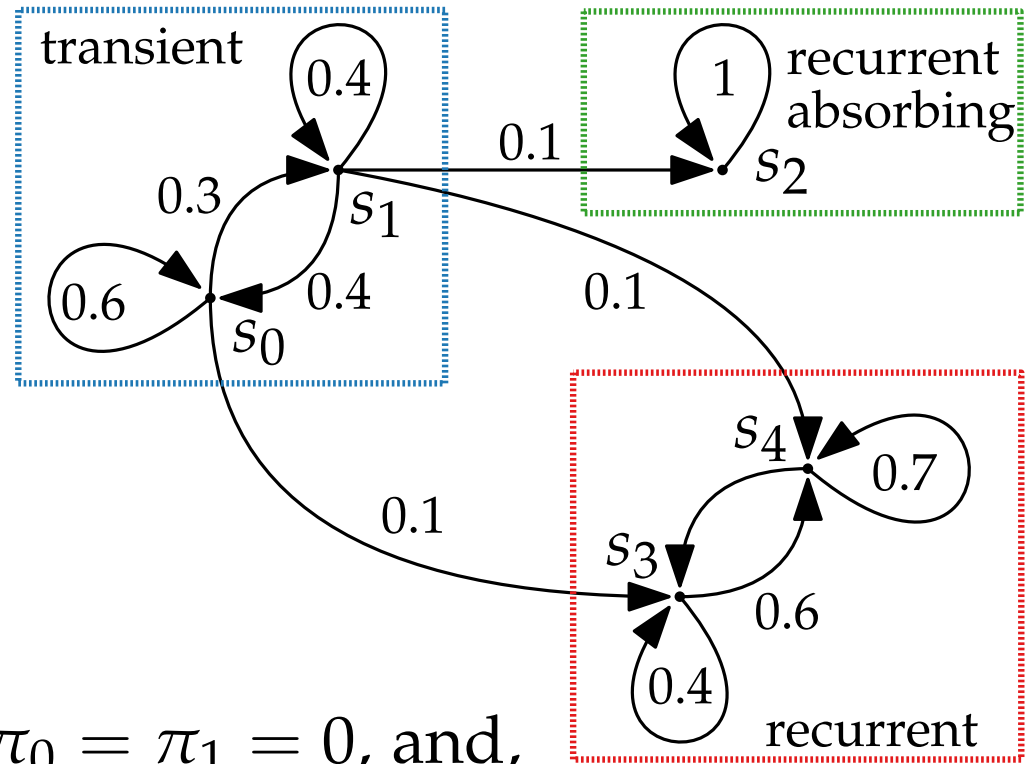
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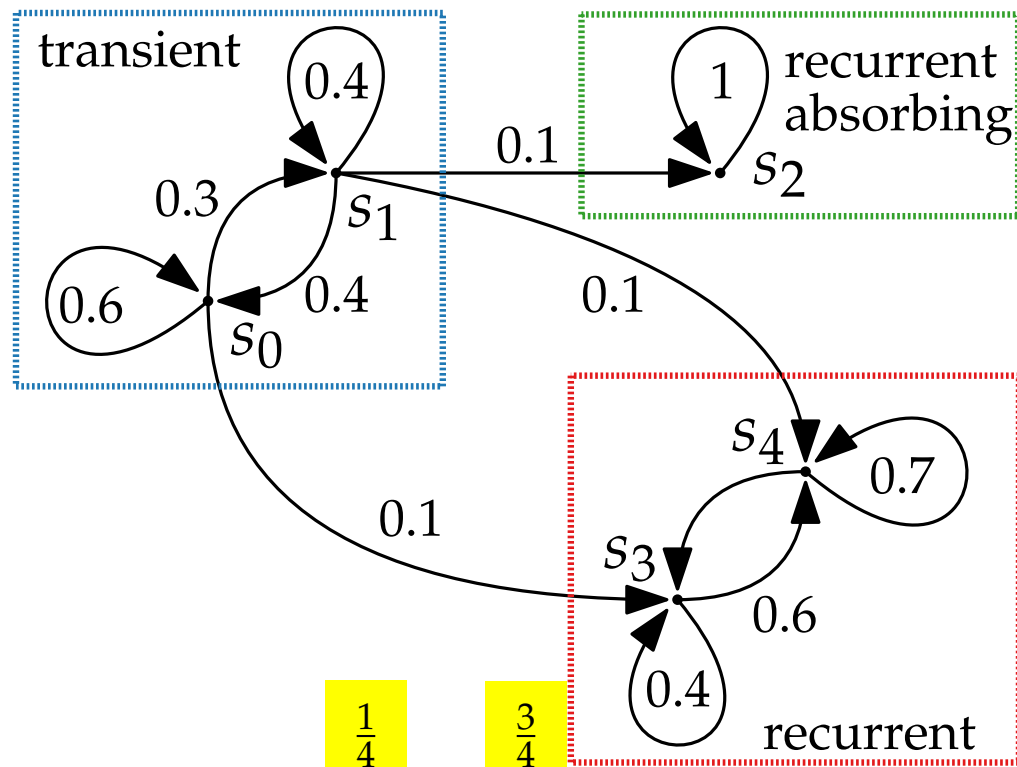
Thus, $\pi = (0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$

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Starting in s_0 :

- Calculate the probability of reaching state (i) s_2 , (ii) s_3 , (iii) s_4 .
- Calculate $\pi = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = (0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$
- On average (long run), how many **times** is (i) s_0 , (ii) s_1 visited?



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c) On average (long run), how many **times** is (i) s_0 , (ii) s_1 visited?

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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

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$$\begin{aligned} \begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix} &= \left(I - \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.4 \end{bmatrix} \right)^{-1} \\ &= \begin{bmatrix} 5 & \frac{5}{2} \\ \frac{10}{3} & \frac{10}{3} \end{bmatrix} \end{aligned}$$

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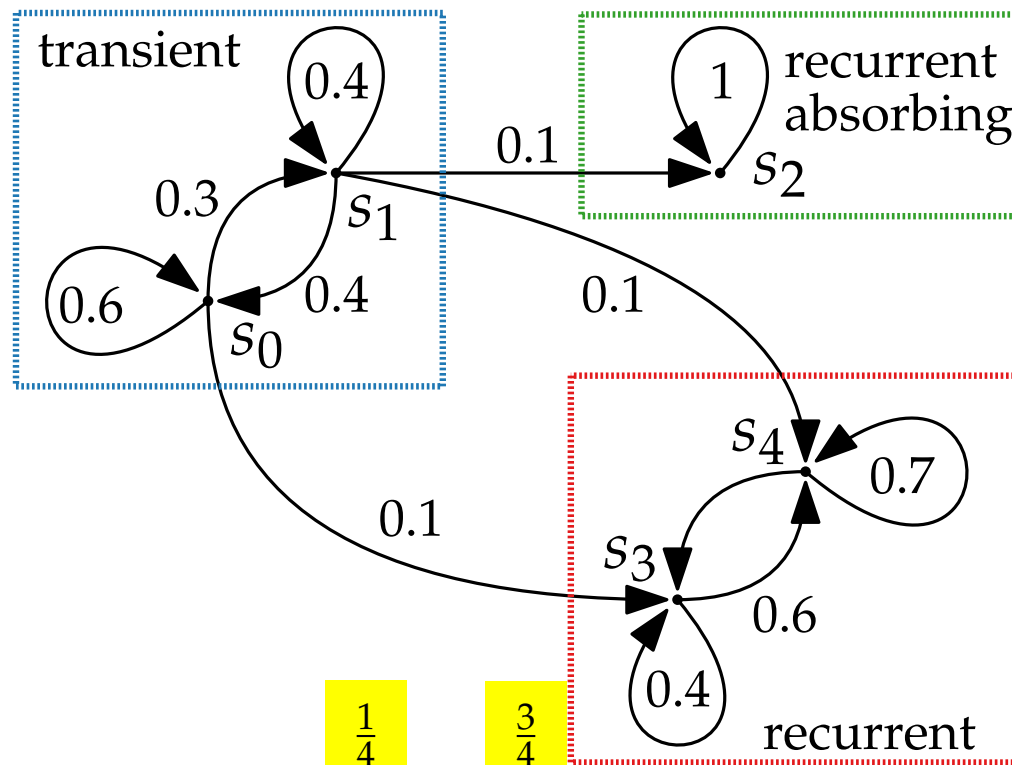
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- On average (long run), how many **times** is (i) s_0 , (ii) s_1 visited?



Starting in s_0 , we stay 5 times in state s_0 and $\frac{5}{2}$ times in state s_1 .